Smartphones

Smartphones are handheld computers with an extensive range of hardware and software functions. They are normally operated using a touch screen (see pp.204–205) that covers their front surface. Smartphones run mobile operating systems and can be customized by downloading and installing apps.

WHAT WAS THE WORLD'S FIRST SMARTPHONE?

IBM's Simon was the first smartphone and was released in 1994. It weighed 18 oz (510 g) and featured a modem for sending and receiving faxes.

What does a smartphone do?

Smartphones combine the features of a telephone and a small computer. They enable communication via a cellular network, Wi-Fi, Bluetooth, and GPS and are equipped with cameras, microphones, loudspeakers, and sensors, while millions of different services are available on app stores. The rise of these powerful, convenient devices has led many specialized devices to become defunct.



Loudspeaker
A miniaturized loudspeaker is built into the phone to provide sound for calls and media. It also enables a speakerphone function for hands-free calling.



Microphone
This allows a smartphone to function as a telephone. It also has a recording function and enables communication with digital assistants.

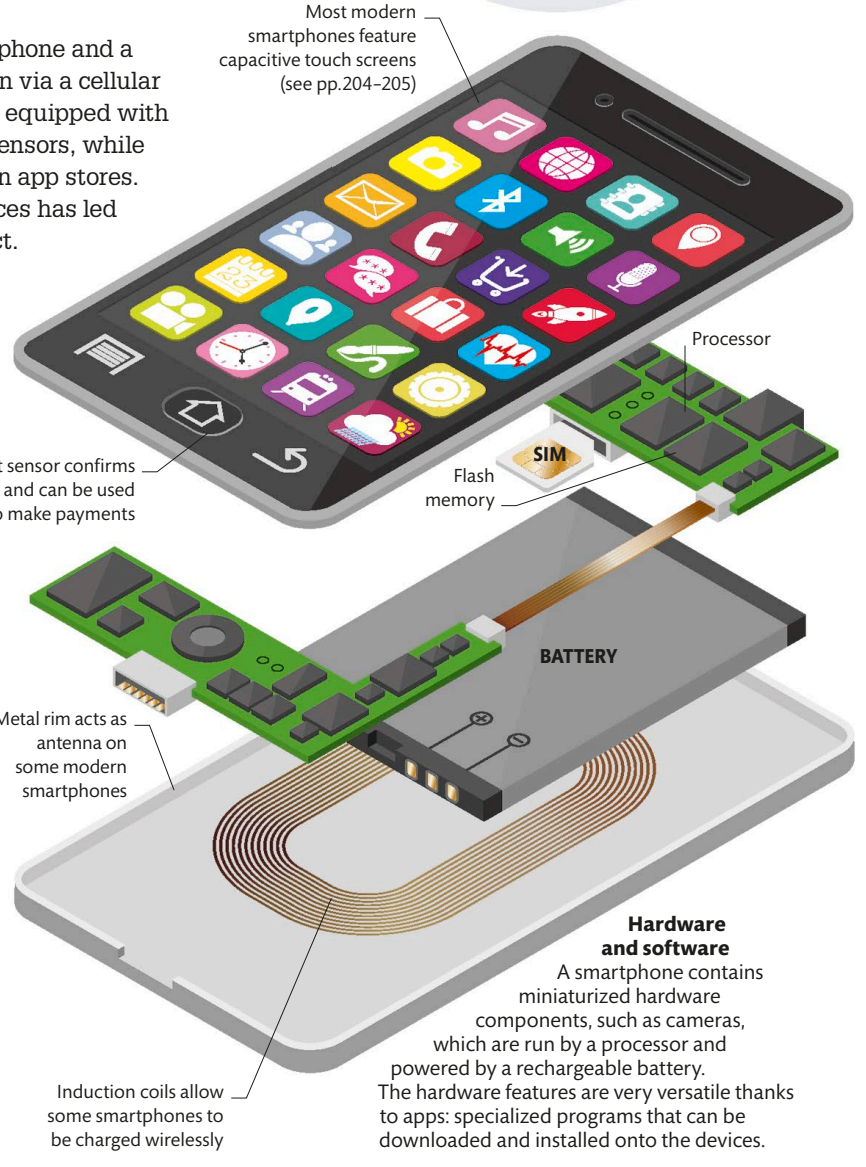
Camera
Virtually all smartphones have small, low-power cameras facing forward and backward. Most have a digital zoom feature and a flash produced by light-emitting diodes (LEDs).



Bluetooth
A Bluetooth chip allows the phone to connect wirelessly to other devices using radio signals. It also enables connection to Bluetooth headsets.



Satnav
A satellite navigation chip connects to a network of satellites in orbit, such as the US Global Positioning System (GPS). The satnav services are accessed through apps.



Hardware and software
A smartphone contains miniaturized hardware components, such as cameras, which are run by a processor and powered by a rechargeable battery. The hardware features are very versatile thanks to apps: specialized programs that can be downloaded and installed onto the devices.